INCH-POUND

MIL-P-48698 (AR)
AMENDMENT 4
11 May 1992

SUPERSEDING
AMENDMENT 3
3 May 1989

MILITARY SPECIFICATION

PROBE AND S&A ASSEMBLY, TOW-2/E6

This Amendment forms a part of Military Specification MIL-P-48698 (AR), dated 27 April 1987, and is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1 Add the following:

"MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

Delete:

DOD-STD-2000 - Soldering Technology - High Quality/High Reliability"

Substitute:

"DOD-STD-2000-4 - General purpose soldering requirements for electrical and electronic equipment."

AMSC N/A FSC 1336

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the current version before use.

* 3.4, delete paragraph in its entirety and substitute the following paragraph:

"3.4 Safety and arming device. The Safety and Arming device (S&A) shall conform to all requirements of MIL-S-14799. The S&A shall be in the safe condition prior to the subsequent assembly of the probe. The red plug shall be evident inside the base terminal board to indicate the piston actuator has not been fired. The rotor shaft shall be in the proper orientation (See Figure 9). The gap between the spring detent and the bottom of the G sensing leaf slot shall not exceed .033 maximum (See Figure 9). Gap that exceeds .066 shall be classified as "critical" defect. Gap greater than .033 but less than or equal to .066 shall be classified as "minor" defect. The piston actuator spacer (Dwg. 9231496) shall be present. No shorting screw shall be present."

3.6, Line 3: Delete "is".

* 3.17, Delete paragraph in its entirety and substitute with:

"3.17 Open circuit test. The probe shall exhibit an open circuit resistance greater than or equal to 1.0 megohms across the tip switch and the ogive switch, and a circuit resistance of greater than or equal to 900k ohms but less than or equal to 30 megohms across the detonator switch in either the stowed or deployed position. Detonator resistance between 100 and 100k ohms shall be classified as "critical" defect. Detonator resistance less than 100 ohms or greater than 100k ohms but less than 900k ohms shall be classified as "major" defect. Detonator resistance greater than 30 megohms shall be considered a "no test"."

* 3.18, Delete paragraph in its entirety and substitute with:

"3.18 Continuity circuit test. The probe shall exhibit a resistance less than or equal to 1.0 ohms for the tip ground circuit, base S&A actuator circuit, ogive switch ground circuit and ogive switch live circuit in either the deployed or stowed position. The stowed probe shall exhibit a resistance of less than or equal to 1.0 ohms for the tip actuator circuit. The tip piston actuator resistance shall be 0.65 $\pm$ 0.15 ohms. Actuator resistance greater than 0.8 ohms shall be classified as "critical" defect. Actuator resistance less than 0.5 ohms shall be classified as "major" defect. The base detonator circuit shall exhibit a resistance of 100k$\pm$10k ohms."

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:37Z
Check the source to verify that this is the currentversion before use.

3.19 Third line, delete specification "MIL-P-70661" and substitute "paragraph 4.5.16".

3.23, Workmanship: Delete paragraph in its entirety and substitute the following:

"3.23 Workmanship.

3.23.1 Metal parts. The TOW/E6 Probe and S&A Assembly and its components shall be fabricated and finished in a thorough, workmanlike manner and shall be free of burrs, chips, sharp edges, unblended radii, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. Wires shall not be damaged during or after assembly. Solder connections shall be in accordance with paragraph 3.20. The cleaning methods used shall not be injurious to any part nor shall parts abe contaminated by the cleaning agents. All required marking shall be clearly legible.

3.23.2 Plastic parts. Parts shall be free from blisters, cracks, foreign inclusions, scratches, voids or any other surface or internal defect which may detrimentally affect serviceability or appearance, and shall be uniform in color and appearance."

3.24, Add new paragraph as follows:

"3.24 Process documentation. Concurrent with submission of the First Article Sample (see paragraph 4.3), the contractor shall submit a written description of the manufacturing process used for the vydax lubrication of the wedge. The manufacturing process shall be reviewed by and must be acceptable to the Government. The contractor's written plan of process and process controls shall be prepared in accordance with Data Item description DI-P-1604 (Tailored) (see 6.9). Proposed changes to the process which produced the accepted First Article Sample shall be submitted to the Government prior to actual implementation of the change. A new First Article Sample may be required for each proposed process change as determined by the Government."

* 3.25, Add the following new paragraph:

"3.25 Leaf spring contact. The leaf spring contact tip in the safe position shall be inspected in accordance with 4.5.17."

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the currentversion before use.

PAGE 6

4.3.1, Add at end of paragraph as follows:

"The First Article Sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packing shall be the same as used in regular production and shall be obtained from the same source of supply. A First Article Sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as the significantly affect product uniformity or quality as determined by the Contracting Officer."

4.3.2, Line 4: Delete "byu" and substitute "by."

PAGE 41

4.3.2.31, Major 114: Delete paragraph in its entirety.

4.3.2.31, Major 115: Change paragraph to the following, "Height of tab straight shear, (015 min.), (see Note 1)".

PAGE 42

4.4.2.31, Minor 208: Delete "3.2" and substitute "3.23".

PAGE 46

4.4.2.33, Major 124: Delete paragraph in its entirety.

Major 125: Change paragraph to the following: "Height of tab straight shear, (.015 min), (see Note 1)."

PAGE 50

4.4.2.35, Minor 211: Delete "3.2" and substitute "3.23."

PAGE 53

4.4.2.37, Minor 206: Delete "3.2" and substitute "3.23."

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the currentversion before use.

PAGE 60

* 4.4.2.44, Delete Major 101 in its entirety.

* Add Major 106 as follows:

"Major	Examination or Test	AQL	Requirement Paragraph	Paragraph Reference
106	Presence of adhesive (see Note 4 of drawing)	100%	3.2	Visual/ app. gage"

PAGE 76 & 77

4.4.2.60: Next higher assembly: Delete "9398046" and substitute "9388046."

* Delete Major 114 in its entirety.

Critical category, delete Critical defect 1 and replace with the following new critical inspections:

"CATEGORY	EXAM/ TEST	CONFORMANCE CRITERIA	REQ. PARA	PARA REF/ " INSP METHOD
Critical	(See note 1)			
1	Modified M114 S&A in safe position	100%	3.4	4.5.1.2a/Gage
2	Tip detonator switch resistance	100%	3.17	4.5.14/STME
3	Tip actuator circuit resistance	100%	3.18	4.5.15/STME

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the currentversion before use.

* Add the following special inspections:

CATEGORY SPECIAL	EXAM TEST	AQL	REQ'D PARA	PARA REF/ INSP METHOD
10	Absence of adhesive sealant (See note 18 on Dwg.)	100%	3.2	Visual
20	Torque on screw - 2 places (See Note 3 on Dwg.)	100%	3.2	Gage
30	Depth of housing support screw <See Note 19 on Dwg.>	100%	3.2	Gage

Delete note 1 and substitute with the following:

"1. In accordance with MIL-A-48078."

PAGE 77a

* 4.4.2.61, Add new paragraph as included.

PAGE 77b

* 4.4.2.62, add new paragraph as included.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the currentversion before use.

QUALITY CONFORMANCE INSPECTION

MIL-P-48698 (AR)
Amendment 4

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.61	Mechanism Assembly		9388005
			NEXT HIGHER ASSEMBLY 9387141

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leaf spring in proper position		100%	3.24	Visual
102	Presence of shoulder washers		100%	3.2	Visual
103	Presence of piston actuator spacer		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.23	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-22 19:37Z
Check the source to verify that this is the current version before use.

(77a)

CLASSIFICATION OF CHARACTERISTICS

MIL-P-48698 (AR)
Amendment 4

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.62	Safety and Arming Device Assembly	1 OF 1	9231500
			NEXT HIGHER ASSEMBLY: N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	(See Note 1)			
1	Position of rotor shaft not horizontal	100%	3.4	Visual/(See Fig. 9)
2	Orientation of G-sensing leaf and detent	100%	3.4	Gage/(See Fig. 9)
Major				
101	Absence of piston actuator	100%	3.4	Visual
102	Absence of safety screw	100%	3.4	Visual
Minor				
201	Improper gap between spring detent and G-sensing leaf.	100%	3.4	Gage/(See Fig. 9)

NOTES:
1. See MIL-A-48078

8 (77-b)
Source: https://assist.dla.mil -- Downloaded: 2026-02-22T19:37Z
Check the source to verify that this is the current version before use.

PAGE 79

Add the following note to the Lot Quality Confirmation Test:

"If only one defect occurs during the Breakaway Strength Test, the same defective unit shall be retested by the contractor for the Breakaway Strength Requirement. Lot acceptance can be granted only when the retest unit meets in full compliance with the test requirements. When retest unit fails to meet the test requirement:

a. Physical Configuration Audit (PCA) on the defective unit shall be performed by the contractor to ensure configuration.

b. Failure analysis report on the defective unit shall be submitted to the Government for review and approval prior to the disposition of the lot."

PAGE 82

4.5.2, After the last sentence add: "(see 6.8)."

4.5.3, After the last sentence add: "(see 6.8)."

PAGE 84

4.5.7, Add to end of paragraph the following sentence:

"A record of actual deployment time for each probe shall be maintained for the lot quality confirmation testing (LQCT)."

PAGE 88

* 4.5.10.2, add the following new paragraph:

"c. Tip assembly. Upon insertion of inert charge assembly (dwg. 9399673), inner tip assembly (dwg. 9399684), and crush ring (dwg. 9399727), assemble the tip assembly (dwg. 9399730) to the probe with 55 + 5 in-lbs torque and cure at ambient for 72 hours minimum before commencing first article or lot quality confirmation testing."

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the current version before use.

PAGE 89-A

Figure 9, add new page as included.

PAGE 90

* 4.5.14.2.a, After the first sentence add the following:

"This test shall be conducted prior to continuity circuit tests described in 4.5.15 (Table V). The test circuit shall not exceed 40 microamperes."

Second line replace "2 milliampers" with "40 microampers for live S&As, or 2 milliampers for inert S&As."

PAGE 90

* Table IV, Delete in its entirety and substitute new Table IV as follows:

TABLE IV - Open circuit test

TEST	FROM (+)	TO (-)	RESISTANCE (ohm)
1. Tip detonator switch	Red cable	Inner tip	$\geq$ 900k and $\leq$ 30 meg
2. Tip switch	Inner tip	Black cable	$\geq$ 1.0 meg
3. Ogive switch	Base contact	ogive (or rivet)	$\geq$ 1.0 meg

4.5.14, delete paragraph in its entirety and substitute with:

"4.5.14 Open circuit tests. Test in accordance with Table IV. Due to the capacitive nature of the circuit, steady state shall be achieved before performing actual measurements. (ref. 3.17)"

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the current version before use.

Table V, Delete in its entirety and substitute the new Table V as follows:

"TABLE V - Continuity Tests

TEST	FROM (+)	TO(-)	RESISTANCE (ohm)
Tip ground circuit	Outer tip	Black cable	≤ 1.0
Ogive switch ground circuit	Ogive (or rivet)	Black cable	≤ 1.0
Base S&A actuator circuit	White/yellow cable	Yellow cable	≤ 1.0
Ogive switch live circuit	Base contact	E10	≤ 1.0
Inner tip circuit	Inner tip	T-lug (S&A)	≤ 1.0
Tip detonator circuit	Pin S&A	Red cable	≤ 1.0
Tip actuator firing circuit	White wire (S&A)	White/black cable	≤ 1.0
Tip piston actuator resistance	White/black cable	black cable	+ .65 ± .15
Base S&A detonator circuit	White/red cable	White/brown cable	= 100k ± 10k

T-lug - terminal lug (dwg. 9388028)
pin - electrical contact stud (dwg. 9231433) "

* 4.5.16, sixth line down after the firing energy, add sentence "in accordance with MIL-P-70661".

* Add the following new paragraph:

"4.5.17, Visual examination for proper leaf spring contact position (See Figure 7 and 8). This examination shall be performed 100 percent at the receiving inspection of the M114 S&A devices."

5. PACKAGING., Delete in its entirety and substitute the following:

"5. PACKAGING

5.1 Preservation and packaging. The probe and its associated components shall be preserved and packaged in accordance with MIL-STD-1169.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing of the probe and its associated components shall be in accordance with MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. The probe and its associated components shall be shipped in the stowed condition in accordance with notes 16 and 17 of Dwg. 9387141."

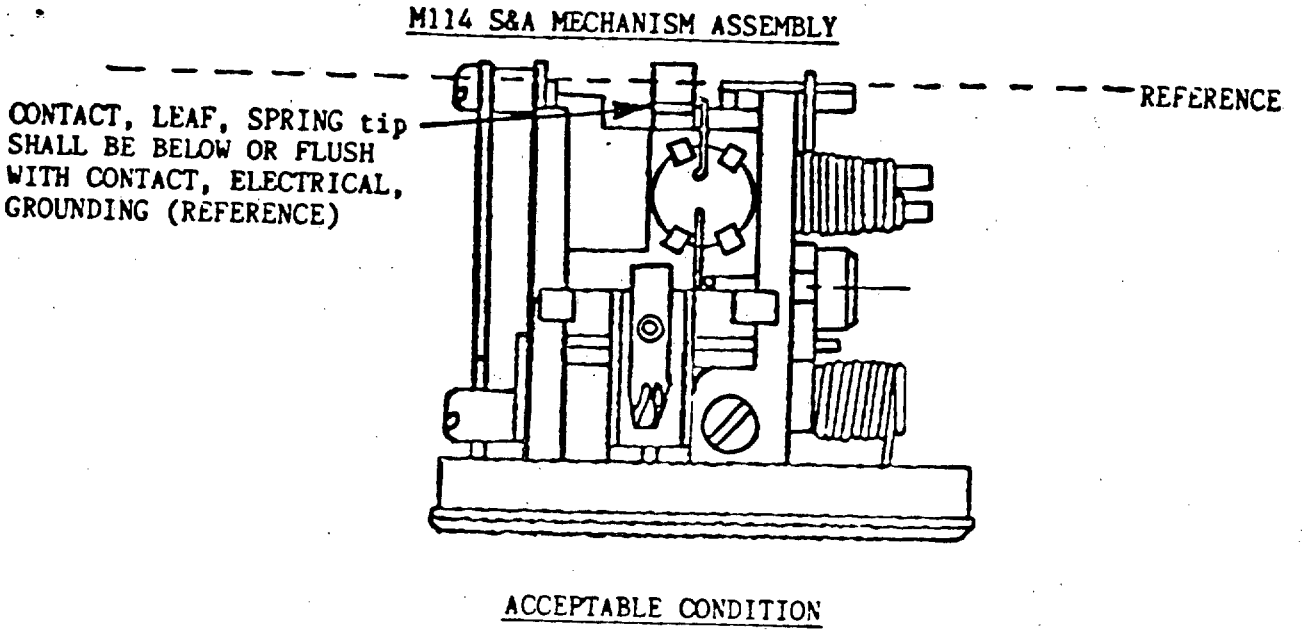
PAGE 91A

* Add Figure 7

PAGE 91B

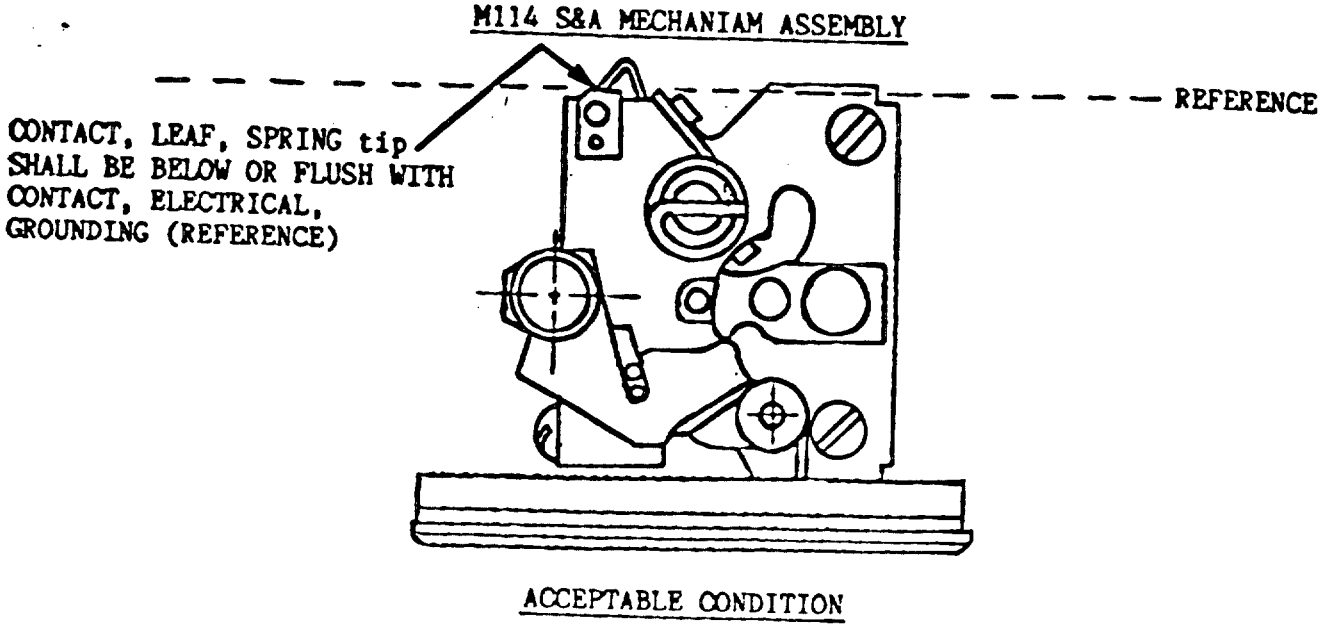
* Add Figure 8

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the current version before use.

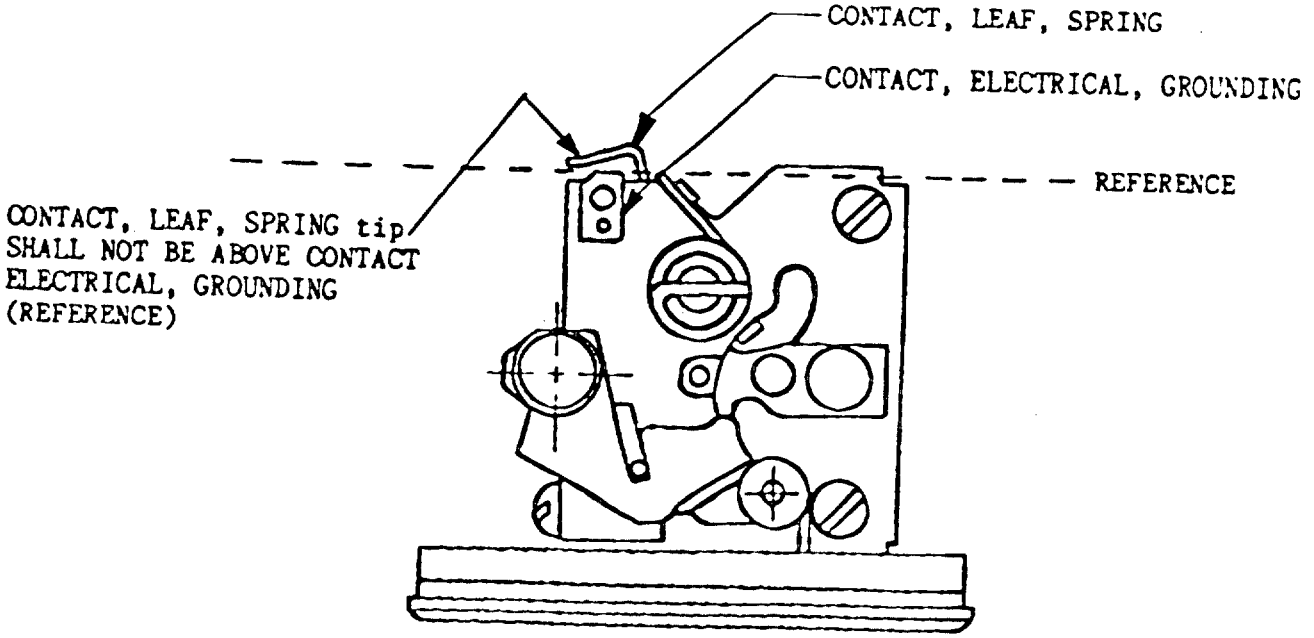


* Figure 7 Acceptance criteria for leaf spring contact (front view)

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the current version before use.



ACCEPTABLE CONDITION



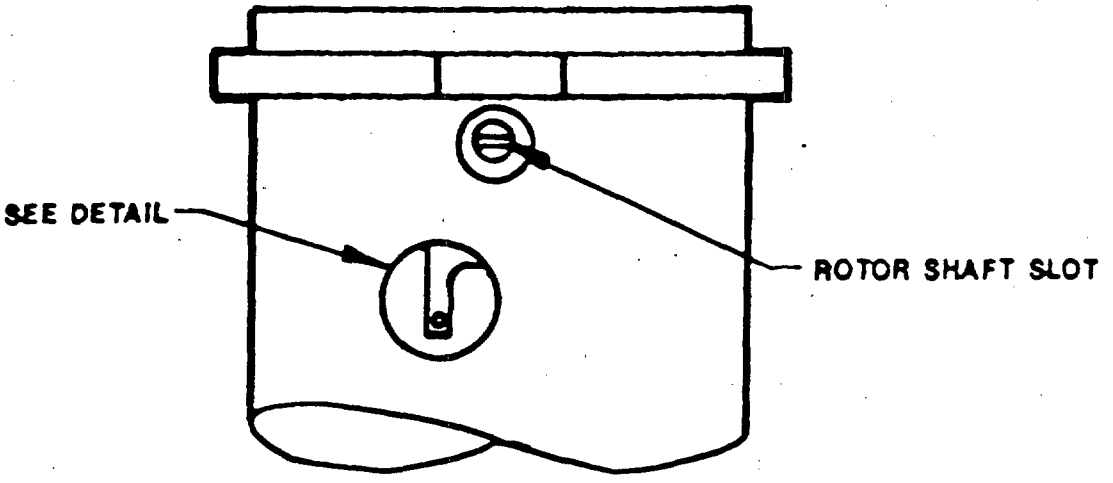
REJECT CONDITION

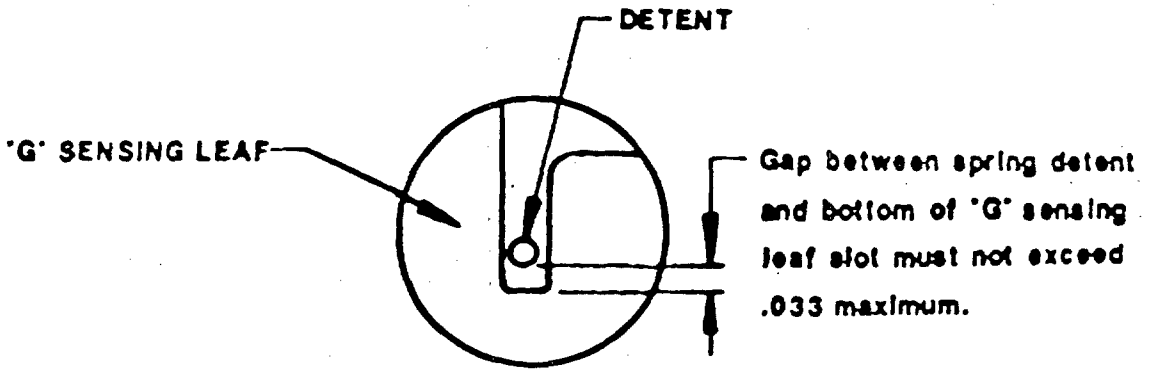
* Figure 8 Acceptance criteria for leaf spring position (side views).

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the current version before use.

SAFETY AND ARMING DEVICE, M114

Safe Condition of Safety and Arming Device (M114)





DETAIL

WARNING

Should the S&A fail to be in the safe condition, an extrame safety hazard shall be considered to exist, and the warhead section disposed of in accordance with acceptable procedures.

FIGURE 9

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

PAGE 92

Add the following:

"3.24	Manufacturing Methods Report	DI-P-1604"

* 6.4, fourth line, delete the following in its entirety:

"3.21	Quality Inspection Test, Demonstration And Evaluation Report	DI-R-1724

PAGE 93

6.8, Add new paragraph as follows:

"6.8 Tip assembly. Upon insertion of the inert charge assembly, inner tip assmbly, and crush ring, apply tip assembly to the probe with 30 ±1 in-lb torque and cure at ambient for 24 hours before commencing first article or lot quality confirmation testing."

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:37Z
Check the source to verify that this is the current version before use.

6.9, Add new paragraph as follows:

"6.9 Process documentation (see 3.24). One copy of the plan shall be prepared and forwarded to the following recipients:

Commander, U.S. Army, ARDEC, ATTN: SMCAR-CCH-A,
Picatinny Arsenal, NJ 07806-5000

and

Commander, U.S. Army, ARDEC, ATTN: SMCAR-QAT-B,
Picatinny Arsenal, NJ 07806-5000

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1336-A627)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.